ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

PURE MATHEMATICS **4027/2**
PAPER 2 2 hours 30 minutes

SPECIMEN PAPER

Additional materials:
Answer booklet
Mathematics data booklet MF7
Mathematical tables
Mathematical instruments
Non programmable Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer paper/answer booklet.

Write your answers on the separate answer booklet provided.

If you use more than one answer booklet, fasten the booklets together.

All working must be clearly shown on the same sheet as the rest of the answer.

Omission of essential working will result in loss of marks.

Decimal answers which are not exact should be given correct to three significant figures unless stated otherwise. Decimal answers in degrees should be given to one decimal place.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.
Non-programmable Electronic calculators or Mathematical tables may be used to evaluate explicit numerical expressions.

This specimen paper consists of 6 printed pages.

Section A [52 marks]

*Answer **all** questions in this section.*

1 Points A, B and C have coordinates (6;4) ; $(-4;2)$ and $(-2;-3)$ respectively.

Find the

(a) coordinates of the midpoint of line AB, [2]

(b) distance between point B and C. [2]

2 Expand $(3x-1)^4$ and simplify. [4]

3 A function $f(x) = 2x^2 + 3x - 1$ is defined for $x \in \mathbb{R}$.

(a) Express $f(x)$ in the form $a(x+b)^2 + c$ where a, b and c are constants. [3]

(b) **(i)** State the coordinates of the minimum point of $f(x)$. [1]

(ii) Find the least value of A for which $f(x)$ is defined for $x \geq A$, has an inverse. [1]

4 Simplify

(a) $8^{\frac{1}{6}} \times 2^{-\frac{1}{2}} \times 2^2$, [2]

(b) $\frac{3}{\sqrt{13}-2}$,

leaving the answer in the form $a\sqrt{b} + c$. [3]

5 Solve the simultaneous equations.

$x + 2y = 4$
$x^2 + 2y^2 = 6$ [5]

6 **(a)** Factorise

$6x^2 - x - 1.$ [2]

(b) Hence or otherwise solve the inequality

$6x^2 - x - 1 > 0.$ [3]

7 Anesu threw a stone to hit a baboon, in the field, with a velocity of *v m/s*. The equation for the motion of the stone, *t*-seconds, after its release is given by

$$v = 2t - 2$$

Find the displacement equation of the stone, after 2 seconds, given that the stone was 11 metres from the point of release. [5]

8 **(a)** Factorise

$9x^2 - 1$. [1]

(b) Express

$\frac{3}{9x^2-1}$ in partial fractions. [5]

9 It is given that

$$(x + B)(Ax^2 + 5x + C) \equiv 2x^3 - x^2 - 13x - 6$$

where A, B and C are constants.

Find the values of A, B and C. [6]

10 **(a)** Given that $x - 2$ is a factor of

$x^3 - 2x^2 + ax + 2$, where a is a constant, find the value of a. [3]

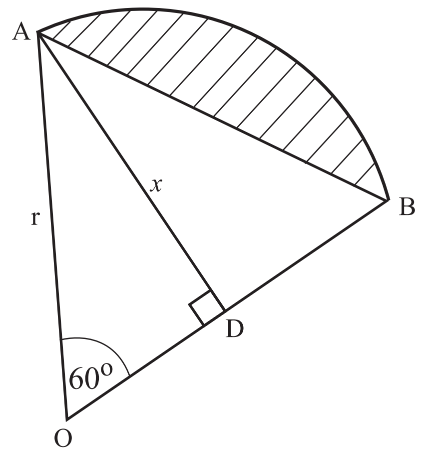
(b) Hence or otherwise factorise

$x^3 - 2x^2 + ax + 2$. [4]

Section B [48 marks]

*Answer any **four** questions in this section. Each question carries 12 marks.*

11 The diagram shows a sector AOB with a point D on OB, such that AD is perpendicular to OB, AD = x cm and the radius of the sector is r cm. Angle AOB = 60° and arc AB = 2π cm.



(a) Convert 60° to radians, leaving the answer in terms of π. [1]

(b) Calculate r. [2]

(c) Calculate the area of the

(i) sector AOB, [2]

(ii) triangle AOB, [2]

(iii) shaded segment. [1]

(d) Show that the length of AD is $3\sqrt{3}$ cm. [2]

(e) Find the length of BD given that the area of triangle ABD is $6\sqrt{3}$ cm^2. [2]

12 Given the equation

$x^3 + 2x - 2 = 0.$

(a) show by calculation that the equation has a root between 0,3 and 0,8. [3]

(b) hence use the bisection method three times to find an approximate value of the root of the equation to 4 decimal places. [9]

13 **(a)** The sixth term of a geometric progression is 16 and the third term is 2.

Find the

(i) first term and the common ratio. [5]

(ii) number of terms of the geometric progression such that the sum is $\frac{127}{2}$. [4]

(b) Given that a sequence is defined by

$u_{n+1} = (4 - u_n)$ and $u_1 = 1$,

find u_2 , u_3 and u_4 . [3]

14 Given that $f(x) = x^3 + 2x^2 + x + 2$,

(a) at the point where $x = -2$, find the

(i) gradient of the curve, [3]

(ii) equation of the tangent to the curve in the form $y = mx + c$. [4]

(b) calculate the co-ordinates of the stationary points of the curve. [5]

15 **(a)** Evaluate $\int_0^2 (x^3 - x)dx$. [3]

(b) Given that

$y = 2x^2 + 1$ for $x \geq 0$,

(i) sketch the graph of

$y = 2x^2 + 1$

for $x \geq 0$. [2]

(ii) shade the region bounded by the curve, the y-axis and the

line $y = 9$. [1]

(iii) calculate the volume generated by rotating the area of the shaded region in **(ii)** about the y-axis through four right angles. Leave your answer in terms of π. [6]

16 Relative to the origin O, the position vectors of points A, B and C are $(1; -4; 6)$, $(0; 1;5)$ and $(-2; 3; 5)$, respectively.

Find

(a) $\left|\overrightarrow{AB}\right|$. [3]

(b) the angle $A\hat{B}C$ correct to the nearest degree. [4]

(c) the unit vector in the direction of $\overrightarrow{AC}$. [3]

(d) the area of triangle ABC correct to two significant figures. [2]